EXPERIMENT 5

Analyze the histogram of an input image based on color levels using OpenCV.

Aim

Analyze the histogram of an input image based on color levels using OpenCV.

Algorithm

• Import OpenCV and Matplotlib.

• Read the input color image.

• Calculate histograms for the blue, green and red channels.

• Plot the channel histograms.

• Save the histogram plot.

Code

```
import cv2
import matplotlib.pyplot as plt

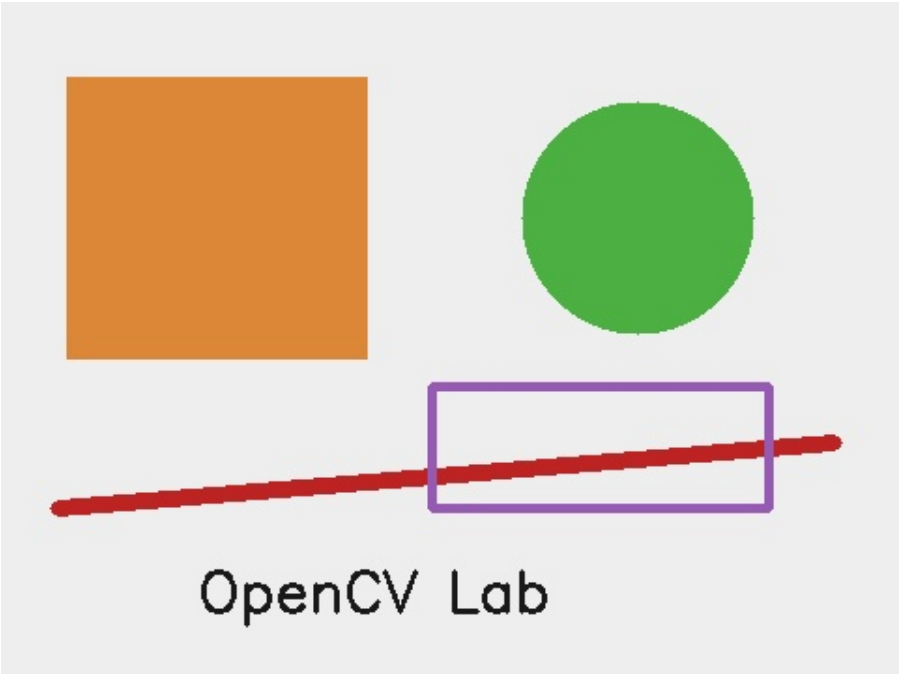
img = cv2.imread("images/input.jpg")
if img is None:
    raise FileNotFoundError("images/input.jpg not found")
for i, color in enumerate(("b", "g", "r")):
    hist = cv2.calcHist([img], [i], None, [256], [0, 256])
    plt.plot(hist, color=color)
plt.xlim([0, 256])
plt.xlabel("Pixel intensity")
plt.ylabel("Frequency")
plt.title("Color Histogram")
plt.savefig("output_images/experiment05_histogram.png")
plt.close()
print("Saved output_images/experiment05_histogram.png")
```

Input Image:

Output Image:

Note: The output image is the plotted color-channel histogram.

Input Image



Output Image

